Predicting SNAP Payment Errors: A Pilot Proposal for New York

Synthetic-data-validated case prioritization for quality-control review

September 29, 2026 · Research whitepaper · Methods and validation artifacts are open-source at github.com/ctrimm/snap-qc-per-experiments

1. Executive summary

We have built and validated a synthetic data generator that reproduces how SNAP payment errors occur in New York's caseload, calibrated against five years of federal quality-control (QC) microdata. Machine-learning models trained *only* on this synthetic data rank real New York cases by error risk above chance (binary AUC 0.66 vs 0.61 for a nationally trained model, consistently across seeds) — proof the synthetic cases carry genuine signal about where errors happen.

The operational finding: a direct expected-dollar ranking model, applied to prioritize QC review, recovers **$6,958 vs $4,574 for random review** at a fixed review budget — **$85 vs $56 per hour of reviewer time**. Reviewing the riskiest half of cases catches 66% of all errors with half the reviews.

We propose a low-risk pilot: train the ranker on New York's own historical QC data, validate on a recent holdout, then run it in shadow alongside existing review. No production eligibility-system changes. No new data collection.

2. The problem

SNAP payment error rates remain stubbornly high nationwide, and every error is either taxpayer money over-issued or a household underpaid. Quality-control review is the corrective mechanism — but reviewer time is finite, and cases are currently selected for review without a data-driven sense of where errors concentrate. The result is expensive review effort spread thinly over a caseload where risk is highly uneven.

New York is a strong candidate for a targeted approach. Our analysis of 3,257 New York QC cases (FY2020–2024) shows the state's benefit-calculation engine matches federal statute at fleet level (94–98% of cases) — New York's errors live in the *human process layer* (verification, timing, documentation), which is precisely where a prediction tool adds value. The pattern is distinctive:

timing errors (cases the agency was not yet required to act on) run at 1.56x the national rate and make up nearly two-thirds of errors, while medical-deduction errors run 1.79x national and standard-utility-allowance errors 1.51x. *Basis note (exp50, October 2026):* the 1.79x figure is pooled FY2020–2024; excluding the partial FY2020 and collapsed FY2021 files, the medical ratio on FY2022–24 is 1.27x — the pandemic years materially inflate this signature.

3. What was built and validated

A New York-conditioned synthetic generator. Starting from a nationally calibrated generator (error-type mix matching reality within ~1 percentage point), we fit New York-specific error-rate offsets and empirical error mixes. The NY-conditioned output reproduces New York's finding mix (L1 0.020), element mix (0.044), and cause mix (0.074), while leaving national output unchanged.

Validated predictive signal — with one honest qualification. Models trained only on NY-conditioned synthetic cases beat baseline and beat a nationally trained model at ranking error vs correct (AUC 0.657 vs 0.611, across all tested seeds), and the advantage survived a fully clean refit excluding the holdout rows. On the element-classification task, however, the NY-conditioned model beats the majority baseline (0.435 vs 0.274) but does *not* beat the nationally trained model (0.468): New York's distinctive signature sits in rare medical/SUA buckets the generator cannot yet place on features. The transferable signal is real, but it lives in the finding and dollar tasks, not in fine-grained element prediction.

A dollar-aware ranking. The set's most important surprise: ranking cases by error probability *alone* performs worse than random at the top slice (0.74x). Error probability and error size are nearly orthogonal in New York's mix — probability chases frequent small-dollar timing errors (median $10–$14) while the dollars live in the income-error tail (newly-employed wages, median $150). The product must therefore be a dollar-aware ranker: here, a direct expected-dollar model won, beating the two-part structure that won in Pennsylvania.

> **Headline pilot economics (simulated on held-out NY cases, top-20% review budget):**
>
> | | Random review | Dollar ranker |
> |---|---|---|
> | Error dollars recovered | $4,574 | **$6,958** |
> | Value per review hour | $56 | **$85** |

Share of all error dollars captured	—	30.4%

Of the $6,958: $4,751 is over-issuance (direct program savings); $2,207 is under-issuance (households underpaid — the correct QC outcome, but a make-whole cost). Savings-only value: $58/hr vs $56/hr.

4. The pilot

What New York provides

- Historical QC case extracts the state already holds (no new data collection, no new PII).
- A review team willing to work a ranked queue for a defined pilot period.

What changes — and what does not

- **Phase 0 (cold start, no NY data needed):** the synthetic-trained ranker scores a sample of cases; rankings are validated against known outcomes.
- **Phase 1:** the dollar ranker is trained on NY's historical QC data and validated on a recent holdout the model never saw.
- **Phase 2 (shadow):** the ranker scores the live review caseload; reviewers work the top slice; outcomes are measured against the baseline selection method.
- **What does not change:** no modifications to the eligibility system, no automated determinations (the model prioritizes review — it never decides a case), no data leaves the state's control.

Success metrics

- Precision in the reviewed slice vs baseline selection.
- Error dollars recovered per reviewer hour vs baseline.
- Reviewer time saved to reach a fixed error-capture target.

A reasonable pilot shape is 8–12 weeks from data handoff to shadow-readout.

5. Limitations and honest caveats

- **Headroom.** The synthetic-trained ranker captures roughly half of the performance a model trained on real NY data is projected to reach (1.52x vs 2.72x lift). The pilot's Phase 1 exists to close exactly this gap.
- **NY-conditioning does not win everywhere.** On fine-grained element prediction the nationally trained model matches or beats the NY-conditioned one; New York's most distinctive errors (medical, SUA) sit in the rare tail the generator's mechanics cannot yet place. Generator mechanics for medical/SUA is the open research lever.
- **Procedural errors are invisible.** Published federal error rates for New York (12.68% FY2023, 14.09% FY2024) run above what case microdata implies (8.54%), in procedural errors the microdata cannot see. This tool targets substantive case errors, not procedural compliance.
- **Timing dominates.** 62% of NY errors are timing variances; the dollar ranker's value comes from the smaller income-error tail. If review priorities shift toward timing compliance rather than dollars, the ranking rule should be re-tuned.
- **Stability.** The signal is year-to-year stable in our tests, but the error-probability baseline should be re-tuned on recent data at pilot start; medical-deduction errors are the one bucket with measurable decay and should be watched.
- **Human judgment stays in the loop.** This is a triage tool for reviewers, not an adjudication engine. Final determinations remain human — by design.

Household-type review burden (exp47, October 2026)

A dollar-maximizing ranker is dollar-efficient and risk-regressive at the same time. Stratifying the holdout by household composition (elderly, disabled non-elderly, single-parent, married with kids, multi-adult with kids, childless adult), elderly and disabled households have the highest error rates (PA: 0.48 and 0.54; NY: 0.34 and 0.42) but the lowest review rates relative to their risk: review-to-risk ratios of **0.22 and 0.33 in PA** — the elderly ratio worse than random allocation (0.42) — and **0.34 and 0.35 in NY**. Childless adults, with the lowest error rates, are reviewed at 1.41x (PA) and 1.45x (NY) their risk.

The mechanism is the dollar objective itself, not a ranking bug: elderly/disabled errors are frequent but small-dollar (medical deductions, timing), while dollars concentrate in income errors — so maximizing dollars per review hour rationally deprioritizes the highest-risk groups. Within each type the ranker is precise (0.62–0.73 precision on reviewed elderly/disabled cases in PA) and selects cases carrying 2–14x the predicted dollars of unselected cases of the same type. The pilot implication: if the goal includes finding underpayments owed to vulnerable households — elderly and disabled households carry high under-issuance exposure — the dollar ranker works against that

goal while optimizing the headline $/hr. A type-aware review budget or a blended dollars-plus-equity objective is the honest next step for Phase 2 design.

6. The ask

We are looking for a New York partner to run this pilot: a data contact who can provide historical QC extracts, and a review team for the shadow phase. We bring the validated methodology, the synthetic cold-start model, and the open-source validation artifacts. All methods, experiments, and results are published openly; nothing about the approach is proprietary or black-box.

Methods, experiment chronicles (60+), and validation code: github.com/ctrimm/snap-qc-per-experiments (private research repo; access on request). This work is conducted in the open-source civic-technology lane. Contact: via the repository.